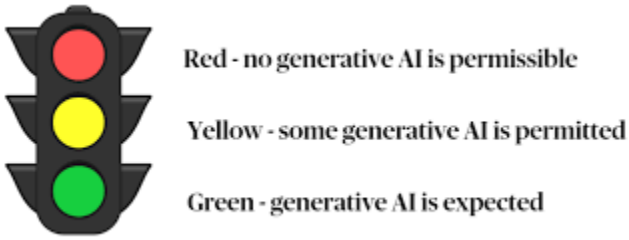
Traffic Light AI Usage Framework

This coursework uses a Traffic Light AI Usage Framework to guide how you may or may not use generative AI tools (e.g., ChatGPT, Copilot, etc.). The colour assigned to this task indicates the level of permitted AI assistance. You are expected to follow these guidelines carefully and ensure all submitted work reflects your own understanding and academic integrity.



Please check the assigned colour for this coursework and adhere to the corresponding requirements outlined below:

🔴 Red – Prohibit to use use gen AI You must not use any generative AI tools at any stage of this assignment. All work must be entirely your own. Any use of AI assistance will be considered a breach of academic integrity.
🟡 Amber – Restricted usage of gen AI Limited use of generative AI is allowed for specific purposes only (e.g., brainstorming ideas, clarifying concepts, or language refinement). AI must not be used to generate substantive content for submission. You must clearly acknowledge any AI use according to the course guidelines.
🟢 Green – Permitted to use gen AI You may use generative AI tools to support your work, including idea generation, drafting, and editing. However, you remain responsible for the accuracy, originality, and integrity of your submission. Proper disclosure of how AI was used is required.

AI Usage Level for This Coursework: 🟡 Amber (Restricted Use)

Guidelines for the Responsible Use of Generative AI (GenAI)

You may use generative AI (GenAI) tools to support learning and development in this assignment in the following ways:

- Brainstorming ideas
- Explaining programming concepts or syntax
- Understanding case study and analysis requirements

However, the following are **not permitted**:

- AI-generated content **must not** be submitted *as-is* as your own work. You are expected to understand, adapt, and significantly modify any AI-generated output.
- GenAI **must not** be used to complete the assessment in full, bypass problem-solving processes, or replace your own reasoning and design decisions.

1.0 COURSEWORK TITLE

Urban Air Quality Analytics: A Multi-Station Study of PM2.5 Pollution in Beijing

COURSEWORK OVERVIEW

For the assignment, you shall explore the application of data analytics techniques to the provided dataset. You must identify and study problems relevant to the dataset, considering the unique properties of the air quality domain, and applying one or more suitable analytical techniques.

Your analysis must be thorough and go beyond the scope of the course. You must incorporate data exploration, manipulation, transformation, and visualisation concepts together with analysis techniques in your solution. Explanations and justifications of each chosen technique must be clearly provided.

Prior to analysis, you must ensure that the data is pre-processed. The assignment should involve several analytical techniques categorised into different criteria, with a detailed exploration of the commands used in each criterion. Findings should be presented, analysed, and justified using appropriate visualisations. A supporting document containing graphs, results, and R programming code must also be submitted.

2.0 OBJECTIVES OF THIS COURSEWORK

This assignment will help you to explore and analyse a real-world dataset, apply data analytics techniques to solve practical problems and construct meaningful information for decision-making. Furthermore, the assignment aims to develop skills in data preparation, visualisation, modelling, and interpretation with the use of R programming, applied here to a real-world environmental and public-health context.

3.0 TYPE

Group Assignment (4 members)

4.0 DATA DESCRIPTION

This dataset contains hourly air-quality and meteorological readings collected from five monitoring stations in Beijing between March 2013 and February 2017, including pollutant concentrations (PM2.5, PM10, SO2, NO2, CO, O3) and meteorological conditions such as temperature, pressure, dew point, rainfall, and wind (refer to **dataset_description.txt**). These attributes provide valuable insight into the environmental and meteorological factors that influence urban air pollution levels.

Objective of the Analysis

The objective is to identify the most significant factors that influence PM2.5 pollution levels and apply predictive models capable of estimating PM2.5 concentration. The findings should support public-health advisories and inform pollution-control decisions, such as identifying high-risk conditions and comparing pollution behaviour across station locations.

Techniques

Appropriate variables should be selected as predictor variables, while the **PM2.5 concentration must be treated as the primary target variable**. The analysis should focus on uncovering relationships between meteorological and pollutant characteristics and PM2.5 concentration through intermediate to advanced concepts, as well as developing models capable of estimating PM2.5 pollution levels.

5.0 GENERAL REQUIREMENTS

- This is a group assignment, requiring equal contribution towards each section.
- Submitted R script(s) must be executed without any runtime or logical errors.
- You **must not use** third-party tools such as Excel, OpenRefine, etc., for data processing. Cleaning and preprocessing must be done in R using scripting.
- You should:
    - Abide to good programming standards such as including comments, variable naming conventions, and proper indentation.
    - Carry out additional research from the Internet to comprehend the knowledge and information on the given dataset when examining the data.

- Additional features must not be redundant and should improve the overall analysis and contribute to the findings.

6.0 DELIVERABLES:

The complete **RScript** (source code) and **Analysis Report** (pdf format) must be submitted to the **APU Learning Management System (Moodle)**.

6.1 RScript (Program Code): *Filename should have you group number*

- Name the file under your group number.
- Start the program with comments of names and TP numbers. For instance:
  # Name1, TP000001
  # Name2, TP000002
  # Name3, TP000003
  # Name4, TP000004
  - For each objective example, provide student id and explain what you want to discover. For example:

    Objective 1: To investigate the relationship between trip characteristics and taxi service performance.

    Analysis 1-1: Which trip characteristics (e.g., trip distance, pickup time, passenger count, payment type) are associated with higher fare amounts?

    Analysis 1-2: Is there a significant correlation between trip characteristics and taxi service performance indicators (e.g., fare amount, trip duration)?

    Analysis 1-3: What external factors (e.g., time of day, weather conditions) interact with trip characteristics to influence taxi service performance?
  - For each extra feature example, give an id and provide the explanation.
    # Extra feature 1
    # description/ explanation about the extra feature and how it help to improve your analysis.

6.2 Documentation (report): *Filename should have you group number*

- A report of approximately 9000 words (excluding appendices and source code screenshots). The report should demonstrate the data analytics flow and comprise of the following sections:

A) Cover Page:

All reports must be prepared with a front cover. The front cover should be presented with the following details:

- Module
- Coursework Title
- Intake
- Students name and id
- Date Assigned (the date the report was handed out).
- Date Completed (the date the report is due to be handed in).

B) Contents:

- Introduction (~800 - 1000 words)
  - ✓ Data Description
  - ✓ Assumptions (if any)
  - ✓ Problem statement (if any)
  - ✓ Hypothesis and objectives
- Data Preparation (~1000 - 1200 words)
  - ✓ Data import and preliminary exploration
  - ✓ Data cleaning and preprocessing (if necessary)
  - ✓ Data validation (if necessary)
  - ✓ Data transformation (if necessary)
- Data Exploration (~600 - 800 words)
  - ✓ General data exploration
    - Distribution of variables
    - Simple visualisations and summary statistics
  - ✓ Key findings
    - General dataset observations

- Basic trends discovered before analysis

- Data Analysis (~4500 - 5500 words)
  - ✓ Each objective (along with student name) must start in a separate page and contains:
    - Analysis Techniques – e.g. descriptive using statistics
    - Screenshot of source code with output / plot.
    - Outline the findings based on the results obtained.
  - ✓ The extra feature explanation must be in a separate page and contains:
    - Screenshot of source code with output/plot.
    - Explain how adding this extra feature can improve the results.
  - ✓ Interpret the results from each analysis
- Conclusion (~300 - 500 words)
  - ✓ Summary of findings
  - ✓ Overall discussion on the findings from all objectives
  - ✓ Limitation and future direction (if any)
  - ✓ State the word count (at the end of page)

C) Workload Matrix

D) References (if any)

- You may source algorithms and information from the Internet or books. Proper referencing of the resources should be evident in the document.
- All references must be made using the **APA (American Psychological Association)** referencing style as shown below:

E) Appendix: AI Declaration Form

F) The font size used in the report **must be 12pt** and the font is **Times New Roman** or **Arial**.

7.0 ASSIGNMENT ASSESSMENT CRITERIA

The assignment assessment consists of 2 major components: Analysis (70%) and finding and discussion (30%). Details of the division for each component are as follows:

<table>
<tr><th colspan="2">Analysis (70%)</th><th colspan="2">Finding and Discussion (30%)</th></tr>
<tr><th>Criteria</th><th>Marks Allocated</th><th>Criteria</th><th>Marks Allocated</th></tr>
<tr><td>Analysis techniques
✓ Approaches used to process, interpret, and extract insights from data.
Report content:
→ Methodology description</td><td>10%</td><td rowspan="2">Finding and Discussion
✓ Project Introduction (description, assumption, hypothesis, objectives)
✓ Conclusion (result finding, discussion, recommendation and future direction)
✓ Structure of the report and references</td><td rowspan="2">30%</td></tr>
<tr><td>Analysis Methods
→ Specific process employed to carry out the analysis (transform raw data into meaningful insights)
[e.g. Exploratory data analysis, Hypothesis test, Descriptive statistics etc]
Report content:
→ RScript or Quarto - code snippet and visualization screenshots with explanation.</td><td>60%</td></tr>
</table>

8.0 DEVELOPMENT TOOLS

The program written for this assignment should be written in R Studio.

9.0 ACADEMIC INTEGRITY

- You are expected to maintain the utmost level of academic integrity during the duration of the course.
- **Plagiarism is a serious offence and will be dealt with according to APU and De Montfort University regulations on plagiarism.**